

Usability Study Report

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Introduction:

Usability testing identifies design problems and understands user behaviours, allowing design decisions to be informed by user evidence (NSW Government, n.d.). This study investigates user interests and needs to inform the development of an educational interface focused on animals. The interface aims to provide an engaging way for children aged five to eight to learn about habitats, diets, adaptations, and survival. The sample consisted of three Year 3 students who were asked about their digital game preferences, usability factors, and features that support engagement. The study utilised four phases of research questions: hook questions involving a familiar zoo scenario, icebreakers about favourite animals, inquiries regarding device accessibility and gaming experience, and game design questions focusing on preferred features, instructions, and feedback.

Data Collection:

Existing knowledge of Animals and Habitats:

Questions were designed to logically hook students using a zoo scenario. Participants mostly recalled giraffes, zebras, and elephants. To identify existing misconceptions, participants were asked what animals eat, where they live, and how they protect themselves. Responses regarding animal protection were mixed; one student mentioned camouflage, another noted physical features like claws and speed, while a third only identified an elephant's trunk. However, all participants shared a common understanding of animal protection through environmental care, specifically highlighting the importance of providing food, water, and safe habitats, which indicates a strong sense of responsibility.

User Interests and Learning Preferences:

The interview identified a range of interests and learning preferences relating to animals.

Area	Participant 1	Participant 2	Participant 3
Animal Preference	"My Favourite animal is a Panda."	"It is probably a Panther."	"My favourite animal is a Giraffe."
Habitat and Physical characteristics	"They live in places with bamboo. They live in a forest."	"It is a big black cat and it lives in jungle environments."	"A Giraffe's habitat is a plain land, with grass and trees."

Animal Needs and Wellbeing	"They need food and a proper habitat for them to live in."	"Access to food and not being hunted."	"They need leaves from trees, food and water."
Learning Interests	"How they adapt to their surroundings."	"How they've adapted to their surroundings, and why they are black opposed to other big cats."	"I like learning about how they look because their appearance looks amazing. Because they look very different from any other animal."
Animal and Habitat Conservation	"Make sure we don't cut down forests, because that is where they live."	"Protect it because jungles seem to be disappearing."	"Give them food and water and separate cages from predators so they cannot come."
Future learning Interests	"I wonder more about Wombats."	"I'd like to learn more about birds. Like Native Australian Birds."	"Yes, it's a Rhino. I want to learn about a Rhino because they look very nice and I don't know much about them."

User Experience and Accessibility Needs:

These questions targeted users' comfort zones and accessibility preferences. Students prefer simple features that make game comprehension and point-scoring easy. Participants favoured simple controls like tapping, clicking, and dragging over traditional keyboards and controllers. While visual instructions were generally preferred, one respondent noted that simple text instructions are sometimes necessary as purely visual cues can be confusing. The preference for mobile or tablet devices highlights the need for touch-optimised interactions. Finally, participants exhibited specific knowledge gaps and confusion regarding the habitats and food patterns of various animals.

Interface Design and Interaction Preferences:

This phase explored visual and interactive preferences. Participants conceptualised dangerous environments as having darker colors or spiky shapes, whereas safe habitats should feature bright, vibrant colors and friendly characters. When asked about task completion feedback, participants strongly desired immediate on-screen rewards: Participant 1 suggested a high score system, Participant 2 requested shiny gold stars and happy sound effects, and Participant 3 preferred in-game collectibles. All participants expressed a strong preference for interactive games over stories or videos, noting that actively helping animals kept them highly engaged.

Data Analysis:

The interview responses reveal a significant curiosity about animals, providing a strong rationale for the game plan. Common interests in giraffes, zebras, and pandas offer an accessible entry point. However, students demonstrated uneven understanding and confusion regarding animal diets, habitats, and survival mechanisms. Consequently, the game design should incorporate stages where users match animals to their correct food and habitats, alongside levels requiring decisions about survival processes. Given the students' empathy, the design will include scenarios where player choices actively improve the characters' lives. This aligns with Ouariachi et al. (2017), who assert that involving players in decision-making strengthens interactive learning.

The preference for mobile tapping and dragging strongly recommends touch-based accessibility. The mixed desire for visual and worded instructions indicates that a single instructional mode may not meet all learners' needs.

Participants exhibited common interests in adaptations, physical characteristics, and wellbeing, with specific interest in how animals adapt and look. They also wanted to learn about unfamiliar species like wombats, rhinos, and native Australian birds. Across all responses, users correctly identified food, water, and protection from predators as vital to wellbeing. These findings dictate that the interface must incorporate a variety of familiar and unfamiliar species, offering educational content on adaptations, diets, and conservation to support existing interests and provide new learning opportunities.

The user preference for distinct visual contrasts such as dark or spiky elements for danger, and bright environments for safety, which aligns with children's intuitive emotional responses. Leveraging these distinct visual metaphors reduces cognitive load, allowing young users to instantly recognise the game state without reading complex text.

Furthermore, the unanimous desire for immediate rewards highlights the critical role of positive reinforcement. Implementing immediate visual and auditory celebrations satisfies Nielsen's (1994) heuristic of "Visibility of System Status," ensuring continuous progress updates that sustain engagement and motivation. Finally, the preference for actively helping animals dictates direct manipulation mechanics; allowing users to actively alter the digital environment provides a sense of agency that translates to interactive gameplay.

Conclusion:

The analysis of the three Year 3 students offers a methodological tool for designing a browser game grounded in users' existing knowledge. While centered on environmental science, the design thoughtfully aims to enhance critical thinking and environmental responsibility. By carefully considering users' interests, knowledge gaps, and accessibility preferences, the game will foster concepts of biodiversity and conservation. Ultimately, this usability study provides valuable user-centered design data, ensuring the final content, format, and interactions effectively meet the target demographic's needs.

References

Nielsen, J. (1994). Enhancing the explanatory power of usability heuristics. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 152–158.

<http://dl.acm.org/doi/10.1145/191666.191729>

NSW Government. (n.d.). *Usability testing*. Digital NSW.

<https://www.digital.nsw.gov.au/delivery/digital-service-toolkit/resources/user-research-methods/usability-testing>

Ouariachi, T., Olvera-Lobo, M. D., & Gutiérrez-Pérez, J. (2017). Analysis of online climate change games: Exploring opportunities. *Revista Electrónica de Investigación*

Educativa, 19(3), 101–114. <https://doi.org/10.24320/redie.2017.19.3.1298>